

Mini-Whiteboard Reference: Lesson 01, Square and Cube Roots

Routine: Boards down → work independently → show in 3, 2, 1 → do not erase → scan all boards.

Threshold: about 80% correct (e.g. 22 of 28 boards).

Round 1: exact roots (15:00–19:00, slides 11–12)

Purpose: check SC1 (exact roots) and SC2 (checking by squaring/cubing) before WE DO.

#	Question	Answer	Purpose / what an error tells you
R1.1	$\sqrt{9}$	3	Very easy entry. 4.5 = halving.
R1.2	$\sqrt{100}$	10	Easy. 50 = halving.
R1.3	$\sqrt[3]{64}$	4	Similar to Example 1. 8 = used square root.
R1.4	Square, area 81 cm ² : side?	9 cm	Variation (context). Missing units or 40.5.
R1.5	True/false: $\sqrt{36} = 18$	False, 6	Deliberate misconception (halving). Check $18^2 = 324$.

If below threshold: 2–3 minute reteach, then re-check

Reteach: “What number times *itself* gives 49?” $7 \times 7 = 49$, so $\sqrt{49} = 7$; say the check aloud. “What number times itself times itself gives 8?” $2 \times 2 \times 2 = 8$, so $\sqrt[3]{8} = 2$. Draw the 7×7 grid if halving persists.

Re-check on boards: $\sqrt{121}$ (11) and $\sqrt[3]{27}$ (3). Time comes from YOU DO (target F1–F8, C1–C3).

Round 2: estimating (27:00–31:00, slides 16–17)

Purpose: check SC3 (bounding, closer to) and SC4 (1 d.p.) before YOU DO.

#	Question	Answer	Purpose / what an error tells you
R2.1	$\sqrt{20}$: between which two whole numbers?	4 and 5	Very easy. 10 and 11 = halving.
R2.2	$\sqrt{45}$: between? closer to?	6, 7; closer to 7	Easy. “Closer to 6” = not comparing 9 and 4.
R2.3	$\sqrt[3]{30}$: between?	3 and 4	Similar (cube). 5 and 6 = used squares.
R2.4	Estimate $\sqrt{10}$ to 1 d.p.	3.2	Variation. $3.1^2 = 9.61$, $3.2^2 = 10.24$: $0.24 < 0.39$.
R2.5	True/false: $\sqrt{65} \approx 8.5$ “because it is between 8 and 9”	False, ≈ 8.1	Deliberate misconception (halfway). 65 is 1 above 64.

If below threshold on R2.1–R2.3: 3-minute reteach, then re-check

Reteach Steps 1–2 only with $\sqrt{27}$: “ $25 < 27 < 36$, so between 5 and 6. $27 - 25 = 2$, $36 - 27 = 9$, so closer to 5.” Show it on a number line from 5 to 6.

Re-check on boards: $\sqrt{50}$ (between 7 and 8, closer to 7) and $\sqrt[3]{10}$ (between 2 and 3, closer to 2).

If only R2.4–R2.5 are weak: start YOU DO and pull a small group at 34:00 to work C3 together.